

Scale Yes, Shape Yes, Sign No: A Sixth Theorem Closing the Finite-Time/Non-Stationary Corner of Covariant Modified Inertia

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Abstract

A five-theorem chain (Zenodo 10.5281/zenodo.21016309, 21139029, 21148494, 21152331) closes covariant modified inertia (MI) as a realization of MOND — under one standing assumption: **stationarity** of the environment's state. The uncovered corner is the natural next question, and it is also the mechanism class of a concurrent proposal (M.J. Luo, arXiv:2602.14515, Feb 2026): can a **non-stationary** bath state, or a **finite-time/transient** response, produce a persistent inertia *reduction* ($\delta m < 0$, the MOND sign) in the galactic band without unexplained power? We close it in three legs, each adversarially verified. **(i)** For linear coupling to a free field, the response kernel is the field commutator — a state-independent c-number at *all finite times*, for *all* states (including non-Gaussian; verified exactly in Fock and cat states) and all non-stationary linear dynamics (time-dependent couplings and parametric/FRW-type frequency drive); in-band parametric kernel-engineering is separately dead by a $\times 30\text{--}36$ pump-rate deficit. **(ii)** For structured (nonlinear) baths, transient gain is *real* — we exhibit it two independent ways — but it persists only for $\min(T_1, 1/\Gamma_2, 1/\text{detuning})$; in the deep-MOND band both its one-shot preparation ($\times \sim 3 \times 10^{11}$ occupancy shortfall) and its regeneration (coherence discount over population inversion is exactly $\leq \times 2$, and the redshift-drift dephasing floor makes coherence $\sim 5 \times$ worse) exceed the universe's non-stationarity budget by ~ 10 orders, while resonant band coverage forces a ~ 400 -channel comb whose phase-random component over-heats disks $\times 342\text{--}788$. **(iii)** Finite-time spectral broadening of the vacuum — Luo's effect — is real and we reconstruct it: the windowed de Sitter-Unruh response obeys an exact quadrature $a_{\text{eff}}^2 \approx a^2 + H^2 + \kappa^2 c^2/T^2$, $\kappa = 1.746$, whose ΔT -mapping shape is *exactly* this framework's interpolation $v(y) = \sqrt{(1+1/y)}$ (sympy identity, $a_0 = 2a_\Lambda$). But Wightman positivity locks the sign for **every** window and **every** trajectory: $F(\omega) \geq 0$ with $F(-\omega) - F(\omega) = (\omega/2\pi) \int \chi^2$, so no transient protocol inverts populations (adversarial maximum $p_e/p_g = 0.54$ for acceleration bursts with parametric pumping), the dynamical dressing stays $\delta m \geq 0$, and the effective scale is *floored* at $cH = Z \cdot a_0$ — the framework's $a_0 = cH_\Lambda/Z$ sits a factor $Z = 5.79$ below the floor, unreachable from the transient side. The genuinely non-stationary scale the mechanism does produce is window-locked to the orbit, $a_T = \kappa c v/R$ — velocity-dependent ($\kappa c/v \approx 1,700\text{--}17,000$), 2–5 orders above a_0 throughout disks — which yields a mechanism-level discriminator: a window-generated interpolation scale would migrate the radial-acceleration-relation transition with c/v across galaxy populations, and SPARC shows it does not. Conclusion: finite-time de Sitter-Unruh physics supplies the *scale window* and the exact interpolation *shape*, and cannot supply the *sign*. The pentad is a hexad. One residual door is quantified, not closed:

one-shot transients contribute $\sim 1.6\%$ per 10 orbits in dynamically *unsettled* systems — nothing for the RAR, but landing exactly on the framework’s live non-adiabatic σ -spread front.

1. The question, and why it is the last theory corner

Theorems I–V close covariant MI from three sides: locality (Ostrogradsky), field-mediation (Cassini/slip), and the environment route — where the **sign/state clause** (Theorem III) proved that passive/KMS baths only *add* inertia, the **frequency law** (Theorem IV) proved in-band softening from a stationary ghost-free bath requires a sub-drive pole, and the **gas clamp** (Theorem V) destroyed the one four-theorem-compliant construction. Every one of those proofs assumes the bath state (or kernel) is **stationary**.

That assumption is not innocent. Lasers exist: a *non-stationary* medium amplifies. The real universe is not stationary: $H(t)$ drifts. And a particle on a galactic orbit is not probed for infinite time: its acceleration vector rotates at $\Omega = v/R$, so the physically relevant response is a **finite-window** response — precisely the generalization Luo (arXiv:2602.14515) proposes, where short-time acceleration broadens spectra beyond the stationary Unruh result and the broadening is read as the MOND interpolation.

So the corner is sharp: **transient or non-stationary physics, in the galactic band** ($\Omega \in [3.2 \times 10^{-17}, 1.9 \times 10^{-14}]$ rad/s; **10-orbit persistence** = 10^8 – 10^{10} yr, up to 3.5 Hubble times), **with the universe’s actual non-stationarity as the only free power source**. Throughout, both coefficient footings are carried: canonical $a_0 = 9.36 \times 10^{-11}$ m/s² (cH_Λ/Z , ρ_{DE}) and alternate 1.13×10^{-10} (ρ_{total}/cH_0); every verdict below is footing-stable (spreads $\leq \times 1.10$ on budgets; 20.7% on H).

2. Theorem VI (statement)

Theorem VI (finite-time/non-stationary closure). For a worldline coupled to any environment: **(i)** if the coupling is linear and the environment a free (quadratic) field, the response kernel (dissipation + mass renormalization) equals the field commutator — a state-independent c-number at all finite times, for all states and all non-stationary linear dynamics. No state preparation, transient probing, or parametric cosmic driving yields $\delta m < 0$; the in-band parametric escape fails by $\times 30$ – 36 on both footings. **(ii)** if the environment is structured/nonlinear, transient $\delta m < 0$ exists but persists only for $\min(T_1, 1/\Gamma_2, 1/|\omega_0 - \Omega|)$; in the deep-MOND band its one-shot preparation is short by $\sim 3 \times 10^{11}$ in occupancy and its regeneration by ~ 10 orders in power (coherence discount $\leq \times 2$ exactly; redshift-drift dephasing floor $\Gamma_\varphi \approx 15 H_0$ makes coherence $\sim 5 \times$ worse than inversion), while resonant band coverage requires a ~ 405 -channel comb whose phase-random component over-heats disks $\times 342$ – 788 . **(iii)** finite-time spectral broadening of the vacuum is real and Deser–Levin-shaped, but positivity-locked: for every window and every trajectory $F(\omega) \geq 0$ and $F(-\omega) - F(\omega) = (\omega/2\pi) \int \chi^2$, so populations never invert, the dressing stays $\delta m \geq 0$, and every window’s effective scale obeys $a_{eff} \geq cH = Z \cdot a_0$. The broadening supplies the interpolation *shape* (exactly $v(y) = \sqrt{(1+1/$

y)) and a *scale window*, never the *sign*; and its genuinely non-stationary transition scale is velocity-dependent ($\kappa c/v$), not universal.

Scale yes, shape yes — sign, still no.

3. Leg (i): free fields are state-blind at all times

The stationary state-blindness of free fields is an old friend; the finite-time, non-stationary version had to be checked, because “transient” is exactly where intuition expects loopholes.

Operator identity (symbolic). Solving the driven bath mode exactly (Heisenberg/Duhamel, sympy) for an *arbitrary* operator source and *time-dependent* couplings: the influence kernels split into a response part built solely from the commutator — a c-number for a free field, hence identical in every state — and a noise part where all state dependence (including genuinely non-stationary oscillating-squeezed covariances) lives. No stationarity is invoked anywhere. Because it is an operator identity, it covers non-Gaussian states too.

Numerics. An $N=200$ -mode ohmic bath, evolved exactly: the kick response across vacuum / thermal ($\bar{n}=2$) / squeezed ($r=1$) / oscillating-squeezed / coherent states agrees to 2.4×10^{-13} while the noise kernels differ by $\times 5$ and oscillate in time. Adversarial verification replaced one numerically-tautological leg of this test with a direct Fock-space computation of $\langle [x(t), x(s)] \rangle$ in genuinely non-Gaussian states (Fock $n=3$, even cat): spread 9.8×10^{-16} against the analytic $-i \sin(\omega \Delta t) / \omega$.

The honest transient, kept. A sudden coupling quench *mimics* $\mu < 1$ spectacularly — $\mu(0^+) = -62.5$ in our units — and this is exactly the class of effect a short-time argument gestures at. It dies in 1.02 drive periods, decays as $\sim 1/t$ (11.4% of the *positive* asymptotic shift remains at 10 periods), is suppressed $\times 6 \times 10^7$ by adiabatic switch-on, is state-blind anyway, and the asymptote is $\mu_\infty = 1.116 > 1$: anti-MOND.

The parametric escape, closed by budget. One route evades state-engineering entirely: drive the bath *Hamiltonian* (FRW-like $\omega_k(t)$) and engineer the kernel itself. Verified: the kernel is parametrically amplified ($\times 2.4$ achieved) while remaining a c-number — but in-band amplification needs pump rate $2\Omega \geq 6.5 \times 10^{-17} \text{ s}^{-1}$, and the universe supplies $H = 1.8\text{--}2.2 \times 10^{-18} \text{ s}^{-1}$: deficit **$\times 35.9$ (canonical) / $\times 29.7$ (alternate)**. Adiabatic kernel corrections are bounded at $(H/\Omega_{\min})^2 \approx 3\text{--}5 \times 10^{-3}$.

Free fields are therefore closed *unconditionally*: the dS vacuum plus free fields cannot give a state- or transient-engineered MOND sign. Any surviving door must live in structured/nonlinear environments — where the boundary was confirmed explicitly (anharmonic bath or nonlinear coupling makes the kernel state-dependent: spread 1.5×10^{-1} vs 3×10^{-16} harmonic control).

4. Leg (ii): transient gain is real, and its bill is the same wall

We are not in the business of manufacturing deficits: **transient gain exists**, and we exhibit it two independent ways. An inverted two-level unit gives $\delta m(0) = -2g^2\omega_0|w|/(\Omega^2(\omega_0^2 - \Omega^2)) < 0$ (confirmed symbolically via Kubo and by dissipator-free RK4 propagation: $\delta m(\text{ground}) = +0.499$, $\delta m(\text{inverted}) = -0.499$ against analytic 0.505); resonant phase-locked coherence gives $\delta m < 0$ at *first* order in the coupling. Theorem IV is genuinely stationary-only: the transient lives in the

homogeneous sector and needs no sub- Ω pole. The door is ajar — for $\min(T_1, 1/\Gamma_2, 1/|\omega_0 - \Omega|)$; a 5% detuning flips the sign within ~ 10 periods.

Then the accounting, in the only band that matters:

- **Persistence.** Deep-MOND (55%/59% of the band, canonical/alternate footing) needs ≥ 10 orbits — up to 3.5 Hubble times. No one-shot transient survives that at physical dephasing floors; $\sim 99.9\%$ of the on-grid deep-MOND volume is killed by redshift-drift dephasing alone ($\Gamma_\varphi = 14.7/16.1 H_0$), margin $\times 1.1$ at the fast corner.
- **Preparation (the verifier's own open-direction attack, then closed).** Even granting a *free* one-shot: the dS-drift in-band occupancy is $|\beta|^2 = 3.2 \times 10^{-12}$ at band-mid, so $O(1)$ inversion is $\sim 3 \times 10^{11}$ **short even once**; maximal coherence amplitude $\sqrt{|\beta|^2} \leq 1.8 \times 10^{-6}$ even with perfect phase lock.
- **Regeneration.** The inversion pump shortfall $S_{\text{inv}} = 2.9 \times 10^{10}$ is reproduced (band-dependent $1.8 \times 10^5 - 3.8 \times 10^{13}$). The hoped-for coherence discount is **exactly** $\leq \times 2$ (Liouvillian: $\Gamma_2 = \Gamma_1/2 + \Gamma_\varphi$; $|\rho_{ge}| \leq \sqrt{p_e p_g}$ — computed, not assumed), and the drift floor makes coherence $\sim 4.9 - 5.4 \times$ worse: $S_{\text{coh}} \in [1.5 \times 10^{10}, 1.6 \times 10^{11}]$. Footing spread $\times 1.10$.
- **Universality.** Covering the band resonantly needs ~ 405 tuned channels with per-star phase locking; self-locking collapses into the stationary c-number kernel of Theorem III, a single shared phase nets $1/\sqrt{N^*} \sim 3 \times 10^{-6}$ with a 50% anti-MOND sign, and the phase-random remainder heats disks $\times 342 - 788$ in σ_v . In-disk raw-energy headroom: $\times 2.1$.

The transient corner of structured baths thus reduces to the same dS-drift delivery wall the pumped steady state died on — now with the one-shot loophole explicitly priced.

5. Leg (iii): Luo's broadening — reconstruction, one honest win, two locks

We reconstructed the finite-time mechanism from first principles (Gaussian-windowed Unruh-DeWitt response on Rindler and de Sitter worldlines; the Planck spectrum recovered to 2×10^{-4} in the long-window limit).

The effect is real. The windowed response obeys an exact quadrature, $a_{\text{eff}}^2 = 1.003 \cdot a^2 + H_{\text{floor}} + \kappa^2 c^2 / T_{\text{win}}^2$, $\kappa = 1.746$ (max residual 0.1%): short-time acceleration genuinely broadens the spectrum in Deser-Levin form. On this point Luo is right, and it matters.

The honest win. Feeding a_{eff} through the ΔT mapping $\mu = [\sqrt{a^2 + a_\Lambda^2} - a_\Lambda]/a$ and inverting yields — *exactly*, as a sympy identity — $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0)}$ with $a_0 = 2a_\Lambda$. The interpolation this framework postulated in 2025 is the *unique* dS-Unruh ΔT shape. Two groups now arrive at the same functional form from the same physics independently; the shape is no longer a choice.

Lock one: the sign is postulated, not derived. Wightman positivity forces $F(\omega) \geq 0$ for any window, and the exact identity $F(-\omega) - F(\omega) = (\omega/2\pi) \int \chi^2$ — verified to 4×10^{-16} , then re-derived adversarially from the universal Hadamard coincidence structure — is **trajectory-universal and window-universal**. Tested on Luo's actual case (non-uniform acceleration bursts), on sharp top-hat windows, and on parametric/dynamical-Casimir pumps: the maximum achievable p_e/p_g is 0.54 — no inversion, ever, from the vacuum. Fed through the dynamical dressing rather than

the mapping, $\delta m \geq +0.65$ across the whole (a, T_{win}) grid; windowed quantum-Brownian-motion checks confirm Pusz–Woronowicz passivity under arbitrary transient switching. The MOND sign in this construction enters where it entered in Milgrom (1999): in the postulated $\mu \sim \Delta T/T$ map. The broadening dresses the *populations*; it never crosses them.

Lock two: the scale is floored and velocity-dependent. All quadrature coefficients are positive, so $a_{\text{eff}} \geq cH$ for every window and every trajectory. The forced Deser–Levin normalization is $a_{0\text{DL}} = 2cH = 2Za_0 = 11.58 \times$ the framework’s canonical a_0 (both footings; $9.0 \times$ the empirical 1.2×10^{-10}) — and since broadening only *adds* in quadrature, cH is a floor: the framework’s $a_0 = cH_{\Lambda}/Z$ sits a factor $Z = 5.79$ below it, unreachable from the transient side. This sharpens the κ -closure (10.5281/zenodo.20965016): the coefficient is not merely underived by this route — it is *excluded* by it. Meanwhile the genuinely non-stationary scale the window does produce is $a_T = \kappa c v/R$: $a_T/a_{\text{centripetal}} = \kappa c/v \approx 1,700\text{--}17,000$ for $v = 30\text{--}300$ km/s, putting a_T 2–5 orders above a_0 everywhere in disks. The only window that lands on a_0 is $T^* = \kappa c/a_0 \approx 177/147$ Gyr (canonical/alternate) = $10.1/H$ on both footings — the stationary limit Theorems III/IV already govern.

A mechanism-level discriminator falls out. If the interpolation scale were window-generated, the RAR transition would migrate with c/v across galaxy populations — dwarfs ($v \approx 30$ km/s) and massive spirals ($v \approx 300$ km/s) would transition at accelerations an order of magnitude apart. SPARC’s single, population-stable transition (the framework’s own hierarchical fits, e.g. 0.108 dex at $Y=0.70$) already rules this out. Any future finite-window MI proposal must explain why its scale does not carry the c/v fingerprint.

6. What this implies for arXiv:2602.14515

Luo’s finite-time broadening is a real effect, and it independently lands on the same de Sitter–Unruh quadrature this framework postulates — indeed the ΔT -mapping shape is exactly $v(y) = \sqrt{(1+1/y)}$ with $a_0 = 2a_{\Lambda}$, which we regard as convergent support for the *shape*. But (a) the reduced-inertia sign enters through the Milgrom-1999-type mapping, not dynamics: for any window and any trajectory the windowed vacuum response obeys $F(\omega) \geq 0$ and $F(-\omega) - F(\omega) = (\omega/2\pi)\int \chi^2$, so no transient protocol inverts populations and the dynamical dressing stays $\delta m \geq 0$ (vacuum passivity); (b) the genuinely non-stationary scale the mechanism produces is window-locked to the orbit and velocity-dependent ($a_T = \kappa c v/R$, 2–5 orders above a_0 in disks), while every window’s effective scale is floored at cH — the only a_0 -reproducing window being $\sim 10/H$, the stationary limit. The mechanism therefore supplies scale-window and shape, not sign; and it inherits a falsifiable c/v fingerprint that SPARC already disfavors. We would be genuinely interested to be shown wrong on either lock — the named mind-changers are an *interacting-field* windowed calculation exhibiting inversion without violating passivity, or a dynamical (not chosen) lock of the window to $\sim 10/H$.

7. What remains open

Per this program’s standing rule, closures are maps, and maps have edges:

1. **Dynamically unsettled systems (the quantified residual).** One-shot transients contribute $\sim 1.6\%$ per 10 orbits. That is nothing for the RAR — settled disks — but it is *distinctive* where relaxation is recent: the prediction is a MOND-amplitude-vs-dynamical-settledness

correlation, landing exactly on this framework’s live non-adiabatic relational σ -spread front (MI 6–13%, MG exactly 0). Underpowered today; open.

2. **Interacting-field windows.** The positivity locks were proven for free fields pulled back to worldlines (the verifiers’ named boundary). A windowed *interacting*-field calculation showing inversion without violating Pusz–Woronowicz would reopen leg (iii). No candidate exists; the burden is now one specific calculation.
3. **The empirical fronts are untouched:** the fixed-direction $\bar{s}^{\text{TX}}$ ephemeris test (10.5281/zenodo.21137568), Gaia DR4 wide binaries (~Dec 2026), DESI DR3 $w(z)$ (the $a_0(z)$ hostage), high- z BTFR offset sign, and the BIG-SPARC population split (10.5281/zenodo.21140507).

8. Verification and scope

Every quantitative claim above is backed by a committed, runnable script (numpy/sympy, no network), all exit 0, each lane re-run and attacked by an independent adversarial verifier instructed to refute in *both* directions. Verifier corrections are printed beside the results they amend, per this program’s practice: one numerically-tautological state-blindness test replaced with an exact Fock-space computation; one “100% of deep-MOND” claim softened to “~99.9% on-grid, $\times 1.1$ -marginal at the fast corner”; one verifier’s own threshold bug acknowledged and fixed; and two strengthenings added (the parametric-drive extension of leg (i); the cH-floor form of lock two).

Scope: this note closes the finite-time/non-stationary corner of covariant modified inertia. It does **not** claim “modified inertia is impossible” beyond the mapped territory (the interacting-field window of §7.2 is the precise surviving edge), and it does **not** touch the $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ reframing or its live empirical tests, which stand or fall on the data fronts of §7.3. The author retracted all earlier “theory of everything” claims (2026-06-23); this note claims a closure and two locks, not a completion.

Script index

Leg	Scripts (all real_research/reviews/nonstationary_2026_07/, all exit 0)
(i)	laneA_symbolic_state_blindness.py, laneA_numeric_finite_bath.py, laneA_nonlinear_boundary.py, verify_A_freefield_kernel_and_sign.py
(ii)	laneB_transient_gain_model.py, laneB_band_persistence.py, laneB_phase_conspiracy.py, laneB_regen_budget.py, verify_B_structured_adversarial.py
(iii)	laneC_finite_time_response.py, laneC_sign_audit.py, laneC_scale_audit.py, verify_C_luo_adversarial.py

References (chain)

- Zimmerman, C.P., *Scale Without Law*, 10.5281/zenodo.21016309 (Theorems I–II context)
- Zimmerman, C.P., *The Sign Premise Is a State Clause*, 10.5281/zenodo.21139029 (Theorem III)
- Zimmerman, C.P., *The Fourth Horn*, 10.5281/zenodo.21148494 (Theorem IV)
- Zimmerman, C.P., *The Kernel That Builds Its Own Laser*, 10.5281/zenodo.21152331 (Theorem V)
- Zimmerman, C.P., κ -closure, 10.5281/zenodo.20965016 (coefficient non-forcing)
- Luo, M.J., arXiv:2602.14515 (2026); and arXiv:2311.03397 (2023)
- Milgrom, M., Phys. Lett. A 253, 273 (1999) (the ΔT mapping)
- Deser, S. & Levin, O., Class. Quantum Grav. 14, L163 (1997)
- Pusz, W. & Woronowicz, S.L., Commun. Math. Phys. 58, 273 (1978)